

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 12

Tap periods (frames): [96, 106, 87, 104, 51, 73, 152, 72, 104, 141, 104]

Tap amplitudes (pixels): [188.1116689628796, 180.42449944505873, 194.26270872197782, 179.61347388211163, 176.64653973401235, 184.50474248647376, 156.20499351813308, 156.84705926475002, 146.4923206178399, 132.5745073534124, 147.32277488562318, 136.12494260788506]

GPT-5

MDS-UPDRS Item 3.4: Finger Tapping — Scoring Analysis

Given data

- Taps instructed: 10
- Detected taps: 12 (likely includes initial/final partial cycles; standard scoring considers the first 10 complete taps)
- Tap periods (frames) for 11 intervals between 12 peaks: [96, 106, 87, 104, 51, 73, 152, 72, 104, 141, 104]
- Tap amplitudes (pixels) for 12 taps: [188.1, 180.4, 194.3, 179.6, 176.6, 184.5, 156.2, 156.8, 146.5, 132.6, 147.3, 136.1]

I interpret “periods” as inter-tap intervals; irregular spikes or very long periods suggest hesitations/interruptions, and very small periods may reflect brief mis-hits/double taps.

1) Amplitude trend (decrement)

- First 10 amplitudes: 188.1, 180.4, 194.3, 179.6, 176.6, 184.5, 156.2, 156.8, 146.5, 132.6
- Pattern:
 - Taps 1–6: generally high and fairly stable (~176–194)

- From tap 7 onward: clear drop (156 → 157 → 146 → 133)
- Interpretation vs. criteria:
 - “Decrements near the end” (Slight, score 1) matches best: the clear, sustained decrement begins after tap 6, i.e., toward the latter part rather than “midway” (score 2) or “after the 1st tap” (score 3).

2) Speed and rhythm (periods)

- Periods: [96, 106, 87, 104, 51, 73, 152, 72, 104, 141, 104]
- Observations:
 - Most periods cluster roughly ~70–110 frames, suggesting a relatively steady pace overall.
 - Notable outliers: 51 (very short), 152 and 141 (longer hesitations).
 - Counting “interruptions/hesitations”:
 - Conservative interpretation flags the obvious long pauses (152, 141) as hesitations (2 events).
 - The 51-frame short interval could indicate a brief irregular/double-tap-like event (possible interruption), but short intervals can also be natural variability; to avoid overcounting, we treat it cautiously.
 - Net: 1–2 clear hesitations, which fits “one or two interruptions or hesitations” (score 1). It does not meet “3 to 5 interruptions” (score 2) nor “more than 5 or a freeze” (score 3).
- Speed:
 - No strong evidence of mild or moderate overall slowing from the periods alone; variability exists but median cadence appears consistent with brisk tapping. This does not push the score upward on speed grounds.

3) Synthesis vs. rubric

- Interruptions: about 1–2 clear hesitations → score 1 range.
- Amplitude: decrements toward the end, not midway and not from the start → score 1 range.
- No evidence of inability to perform (score 4), nor clear “mild” slowing with 3–5 interruptions (score 2), nor early decrement or moderate slowing/long arrest (score 3).

Final score: 1
